

Problem No. 6 (25 Points):

Solution:

There are three words in this problem which denote direct descendants: grandson, grand grand grandson and grand grand grand grandson. In addition, we can assume that the shoot of rice might be denoted in a similar way. Among the Malagasy words the element *zafy* / *zafim-* / *zafin-* can be found 4 times. Since we have one more word with the component 'rice' – rice field – and neither the component 'rice', nor the component 'field' can be found anywhere else in the problem, we can guess that *zafy* means 'grandson', *zafim-bary* – 'shoot of rice' (lit. 'rice grandson') and *tanim-bary* – 'rice field'. Then the words *zafin-kitrokely* and *zafim-paladia* mean 'grand grand grand grandson' and 'grand grand grand grand grandson' (but we still do not know which one is which), and these meanings in Malagasy are expressed by some other meanings than in English.

Among the remaining words we can find a pair *mahandohalika* – *mahambozona*. The first word is somehow related to the word *lohalika* from the word list and the second one is related to the word *vozona* from the assignment 2. It's quite obvious from these pairs, that in Malagasy some consonant alternations occur when compound words are derived; the variation in the pair *mahan-* / *maham-* is identical to the variation in *zafin-* / *zafim-*. Thus, we can assume that *mahandohalika* means 'one who can get on his knees', *lohalika* – 'knee', and *mahambozona* – 'one who can carry smth on his neck'.

So, we have only two words left – *hafaladia* и *kitrokely* – and two translations – 'ankle' and 'up to the sole'. Since we can find the element *ha-* in the first of them (compare this word with the word *halohalika* in the assignment 2), we can assume that it means 'up to the sole', then the word *kitrokely* means 'ankle'. Now we can see that the words 'grand grand grand grandson' and 'grand grand grand grand grandson' in Malagasy literally look like 'grandson-ankle' and 'grandson-sole'. It's natural to assume that the more distant descendant is compared to a lower part of the body (otherwise it would be problematic to translate the word 'grand grandson' and the word 'grand grand grandson', which can be found in the assignment 3).

Now we can summarize all we know about the consonant alternations which can be found in the problem. If the word ends with -y, it becomes -i- before any other element and the nasal consonant is added to it. This can be *m* (before *b* or *p*) and *n* (before other consonants: *d* or *k* in this problem).

Alternations of the initial consonants (when some element is added to the left):

consonant	after a vowel	after a consonant
k-	?	k
l-	l	d
v-	–	b
?	f	p

It is obvious that there are no alternations of the initial consonant after a vowel (i.e. the root starting with *f-* after *ha-* should also start with *f-* if it occurs without a prefix), but some surprising changes can happen after a consonant. Fortunately, we do not have to find out any new changes in this problem.

Assignment 1:

mahandohalika	one who can get on his knees
zafin-kitrokelygrand	grand grand grandson
zafim-bary	shoot of rice (departing from the stem)
zafim-paladia	grand grand grand grand grandson
tanim-bary	rice field
lohalika	knee
hafaladia	up to the sole
kitrokely	ankle
zafy	grandson
mahambozona	one who can carry smth on his neck

Assignment 2:

tany	field
vozona	neck
halohalika	up to the knee

Assignment 3:

grand grand grandson (grandson-knee)	zafin-dohalika
sole	faladia
up to the ankle	hakitrokely
